Topological Graph-Spectral Machine Learning for Real-Time Surrogate Estimation of Urban Traffic CO_2 Emissions

Thomas Clerc, Pierre Uzarralde, and Alain Faye

***Abstract*—Microscopic traffic simulators such as SUMO and Aimsun are accurate but too slow for iterative planning and real-time control. We propose a surrogate model, based on the spectral properties of the road network, that estimates simulated urban CO_2 emissions in under 6 ms. Each directed road network is represented by two weighted matrices whose rows are normalized and damped by $\beta = 0.95$, so that all eigenvalues lie inside the unit disk. From them we compute spectral descriptors (spectral radius, commutator norm, Kreiss constant, pseudospectral radius, H_2 norm, link sensitivity), which we combine with traffic demand, topology and fleet electrification into a 47-dimensional feature vector. An XGBoost model trained on 349 SUMO simulations from 65 cities reaches $R^2 = 89.39\%$ on emissions per vehicle (5-fold cross-validation: $86.10 \pm 5.12\%$) and a WAPE of 2.74% (unweighted MAPE: 7.28%) on total city emissions, against 13.45% (MAPE: 18.20%) for a GCN. On nine unseen cities, the MAPE is 2.67% on base scenarios and 4.82% over all 18 runs. For a city whose features are precomputed, warm-start inference takes under 6 ms (more than 10^5 times faster than SUMO), while building the features for a new city takes under 132 s.**

***Index Terms*—Urban traffic emissions, Graph-spectral surrogates, Non-normal operators, Kreiss matrix theorem, Pseudospectral stability, Zero-shot transfer learning.**

I. Introduction

URBAN road transportation accounts for more than 24% of global energy-related greenhouse gas emissions and remains a primary contributor to municipal particulate matter ($PM_{10}, PM_{2.5}$) and nitrogen oxides (NO_x) pollution [1], [2]. Accelerating municipal decarbonization requires urban traffic managers to continuously evaluate complex planning interventions, such as low-emission zones (ZFE), dynamic congestion charging, arterial transit lanes, and adaptive traffic signal coordination [3], [4].

Microscopic agent-based traffic simulators such as SUMO (Simulation of Urban MObility) [5] coupled with Handbook Emission Factors for Road Transport (HBEFA 3.3) [6] represent the standard for high-fidelity emissions modeling. By resolving individual car-following dynamics (Krauss model), lane changing, and intersection priority rules at sub-second granularity, microscopic simulators realistically reproduce non-linear queue formation and localized shockwave propagation. However, this microscopic resolution incurs severe computational costs: simulating one peak hour of traffic across a medium-sized metropolis ($m \approx 10^4 - 10^5$ road segments, $n \approx 10^3 - 10^5$ intersections, $N_{\text{veh}} \approx 10^5$ vehicles) routinely requires between 45 minutes and several hours of multi-threaded CPU execution. This computational bottleneck precludes interactive municipal scenario exploration, multi-objective infrastructure optimization, and real-time digital twin actuation.

To overcome this limitation, machine learning surrogates have emerged as a promising alternative. However, prevailing deep learning methodologies face fundamental architectural constraints when applied across disparate urban topologies:

1) **Scale Confounding**: Total emissions are almost proportional to the number of vehicles ($r = 0.9865$). A model trained on total emissions can therefore reach a high R^2 by learning vehicle count alone and ignoring the network structure.
2) **Local Message Passing and Graph Symmetry**: Standard graph neural networks such as GCN [7] use a symmetric normalization of the graph, and even directed models such as DCRNN [8] or T-GCN [9] only pass information between neighboring nodes. On large city graphs, this local message passing can lose global information (over-smoothing). We compare our model with GCN and GAT in Section IV-B (Section IV-B).
3) **Zero-Shot Generalization**: Black-box neural models trained on a single metropolitan area cannot generalize to unobserved urban fabrics without extensive recalibration.

To overcome these fundamental limitations, this paper introduces a physics-grounded topological graph-spectral surrogate framework.

Why Non-Normality in Urban Traffic?: In undirected or symmetric networks, graph operators commute with their transpose ($AA^T = A^TA$), ensuring orthogonal eigenvectors and monotonic perturbation dissipation governed strictly by eigenvalues. In contrast, urban road systems are directed and asymmetric due to one-way street grids, turn prohibitions, and directional lane capacities. Such asymmetric operators are intrinsically non-normal ($AA^T \neq A^TA$). In non-normal dynamical systems, non-orthogonal eigenmodes permit transient perturbation amplification, where localized disturbances (such as an intersection slowdown or queue spillback) can temporarily grow before decaying, mirroring the propagation of stop-and-go congestion shockwaves.

By representing urban road networks as asymmetric non-normal transition operators and extracting their non-normal spectral signatures, our framework maps macro-topology, vehicle demand, and powertrain fleet mix to micro-simulation emissions in inference time under 6 ms.

The main contributions of this work are summarized as follows:

Thomas Clerc is with École Hexagone, France (e-mail: thomas.clerc.pro@gmail.com).

Pierre Uzarralde is a Professor and Director of the Artificial Intelligence Curriculum with École Hexagone, France.

Alain Faye is a Professor with ENSIIE and Laboratoire CEDRIC (CNAM), France, and an Academic Supervisor with École Hexagone, France.

Manuscript received August 15, 2026; revised September 22, 2026.

1) **Non-Normal Matrix Operator Formulation & Dissipative Stability**: We represent each road network by a directed, weighted matrix whose rows are normalized and damped by $\beta = 0.95$, ensuring $\rho(\tilde{A}) = 0.95 < 1$. This guarantees that the H_2 resolvent norm and the discrete Lyapunov equation are well defined, while extracting the one-way link imbalance $\Delta(\tilde{A})$, transient disturbance growth $K(\tilde{A})$, pseudospectral sensitivity $\alpha_\epsilon(\tilde{A})$, and analytical link perturbation sensitivity $S_{\max}$.
2) **47-Dimensional Multi-Physics Descriptor Vector**: We construct a structured 47-dimensional feature space integrating kinematic demand, fleet electrification, physical topology, 20 non-normal spectral invariants, and dimensionless cross-interaction ratios that bridge microscopic vehicular kinematics with macroscopic network topology.
3) **Target Normalization and Empirical Validation**: By formulating the surrogate on the normalized per-vehicle emissions target ($y = \mathrm{CO}_2/N_{\text{veh}}$), we decouple structural topology from vehicle volume. Trained on 349 microscopic SUMO simulations across 65 global cities, our XGBoost surrogate achieves $R^2 = 89.39\%$ on normalized emissions (5-fold CV $R^2 = 86.10 \pm 5.12\%$), translating to a macroscopic city-scale WAPE of 2.74% (unweighted MAPE: 7.28%, RMSE $= 3,680$ kg) on rebuilt total emissions, outperforming GCNs by $+43.47$ percentage points on R^2 (reducing total emission WAPE from 13.45% to 2.74%). We present detailed case studies across six global archetypes and quantify feature importance via TreeSHAP analysis.
4) **Zero-Shot Generalization and Convex Barycentric Diagnostic**: We validate model transferability across nine unseen metropolises, achieving a macroscopic MAPE of 2.67% on representative base scenarios (4.82% over the full 18-run multi-scenario cohort). We introduce a 21-dimensional convex quadratic programming barycentric projection [10] that quantifies topological proximity to the training convex hull, providing an automated confidence metric for zero-shot deployment.
5) **Fast Real-Time Inference for Digital Twins**: We benchmark computational latencies across metropolitan networks with up to $68,000$ raw nodes, demonstrating warm-start inference times under 6 ms (more than 10^5 times faster than SUMO) and cold-start network parsing under 132 s, establishing the surrogate as an operational engine for real-time municipal digital twins.

II. Mathematical Foundation of Non-Normal Graph Operators

A. Asymmetric Directed Graphs and Edge Impedance Matrices

An urban road network is modeled as a directed multigraph $G = (V, E)$, where $V = \{v_1, \dots, v_n\}$ denotes the set of n road intersections (nodes) and $E = \{e_1, \dots, e_m\} \subseteq V \times V$ represents the set of m directed road segments (edges). We keep only the largest strongly connected component, so that every intersection can be reached from every other one. The adjacency matrices are then irreducible [11], [12]. Each directed edge $e = (v_i, v_j) \in E$ is characterized by its physical length $L_{ij} > 0$ (meters), legal free-flow speed limit $v_{\text{free},ij} > 0$ (m/s), number of lanes $c_{ij} \in \mathbb{N}_{\geq 1}$, and storage capacity $C_{ij} = L_{ij} \cdot c_{ij}/l_{\text{veh}}$, representing the number of vehicles that fit on the link when stopped ($l_{\text{veh}} = 7.5$ m per vehicle).

TABLE I
Summary of Primary Mathematical Notations and Physical Analogues

Symbol	Mathematical Definition	Network / Physical Meaning
$G = (V, E)$	Directed multigraph	Urban road network ($n = \lvert V\rvert$ nodes, $m = \lvert E\rvert$ edges)
$A_0 \in \{0,1\}^{n\times n}$	Binary adjacency matrix	Unweighted directed topological connectivity
$A_w \in \mathbb{R}^{n\times n}_{\geq 0}$	Weighted impedance matrix	Free-flow travel-time weighted connectivity
$\tilde{A}_0, \tilde{A}_w$	$\beta D_{\text{out}}^{-1} A$, $\beta = 0.95$	Dissipative, row-stochastic transition operators
$\rho(A_w)$	$\max_{\lambda\in\sigma(A_w)} \lvert\lambda\rvert$	Perron-Frobenius spectral radius (weighted connectivity)
$\Delta(A_0)$	$\lVert A_0 A_0^T - A_0^T A_0\rVert_F$	Commutator norm (spatial imbalance of one-way links)
$K(\tilde{A}_w)$	$\sup_{\lvert z\rvert>1} (\lvert z\rvert - 1)\lVert (zI - \tilde{A}_w)^{-1}\rVert_2$	Kreiss constant (maximum transient disturbance growth)
$\alpha_\epsilon(\tilde{A}_w)$	$\sup\{\lvert z\rvert : z \in \Lambda_\epsilon(\tilde{A}_w)\}$	Pseudospectral radius (eigenvalue sensitivity to perturbation)
$\lVert R\rVert_{H_2}$	$(\sum_{k=0}^{\infty} \lVert \tilde{A}_w^k\rVert_F^2)^{1/2}$	Discrete Hardy H_2 norm (total random disturbance response)
$S_{\max}(\tilde{A}_w)$	$\max_{ij} (w_i v_j / \mathbf{w}^T\mathbf{v})$	Maximum first-order link sensitivity derivative
$\Delta\lambda_{\text{Markov}}$	$1 - \lvert\lambda_2(\tilde{A}_0)\rvert/\beta$	Markov spectral gap (flow mixing and dispersion speed)
$y, \hat{y}$	$\mathrm{CO}_2/N_{\text{veh}}$ (true and predicted)	Normalized emissions per vehicle (kg CO_2/veh or g/veh)
$Y, \hat{Y}$	CO_2 (true and reconstructed)	Total city-scale carbon emissions ($\hat{Y} = \hat{y} \cdot N_{\text{veh}}$, kg)
MAPE	$\frac{1}{M}\sum \lvert\hat{Y}_i - Y_i\rvert/Y_i$	Unweighted Mean Absolute Percentage Error (identical on y and Y)
WAPE	$\sum \lvert\hat{Y}_i - Y_i\rvert / \sum Y_i$	Volume-weighted percentage error across total carbon mass
$d_{\mathcal{H}}$	$\lVert\mathbf{z} - \mathbf{X}_{\text{train}}^T \mathbf{w}^*\rVert_2$	Regularized barycentric distance to training convex hull

We construct two fundamental adjacency matrices governing topological connectivity and kinematic flow impedance:

1) **Binary Adjacency Matrix** $A_0 \in \{0,1\}^{n\times n}$:

$$(A_0)_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E, \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

Parallel roads between the same two intersections are counted as a single link.
2) **Physical Impedance Matrix** $A_w \in \mathbb{R}^{n\times n}_{\geq 0}$:

$$(A_w)_{ij} = \begin{cases} \tau_{ij} \cdot c_{ij} = \dfrac{L_{ij} \cdot c_{ij}}{v_{\text{free},ij}} & \text{if } (v_i, v_j) \in E, \\ 0 & \text{otherwise,} \end{cases} \tag{2}$$

where $\tau_{ij} = L_{ij}/v_{\text{free},ij}$ is the free-flow travel time of the link. The weight $(A_w)_{ij}$ is this travel time multiplied by the number of lanes (in units of lane $\cdot$ s).

B. Perron-Frobenius Spectrum and Topological Asymmetry

Because of one-way streets and turn restrictions, A0 and Aw are in general not symmetric.

For any square matrix $M \in \mathbb{R}^{n\times n}$, we denote its spectrum by $\sigma(M) = \{\lambda \in \mathbb{C} : \det(\lambda I_n - M) = 0\}$, and its spectral radius by:

$$\rho(M) = \max_{\lambda\in\sigma(M)} |\lambda|. \tag{3}$$

For a non-negative irreducible matrix M, the Perron–Frobenius theorem [11], [12] states that:

1) $\rho(M) > 0$ is a simple eigenvalue of M;
2) it has a positive right eigenvector $\mathbf{v}$ and a positive left eigenvector $\mathbf{w}$ ($M\mathbf{v} = \rho(M)\mathbf{v}$, $\mathbf{w}^T M = \rho(M)\mathbf{w}^T$), each unique up to a positive scaling factor;
3) if $0 \leq B \leq M$ and $B \neq M$, then $\rho(B) < \rho(M)$ [11], [12]. In particular, removing a road or lowering its weight strictly decreases ρ.

We use $\rho(A_w)$ as a descriptor of how strongly the network is connected through its heavily weighted roads and loops. We do not claim that it measures traffic capacity, which is better described by max-flow/min-cut and by the macroscopic fundamental diagram [13], [14].

C. Non-Normality, Commutator Norm, and ϵ-Pseudospectra

A matrix M is normal if $MM^T = M^TM$. Symmetric matrices are normal, so a network with only two-way roads is normal. In directed networks, one-way streets can make the matrix non-normal. Asymmetry alone is not enough, though: a cyclic permutation matrix is asymmetric but normal. We therefore measure non-normality directly.

To quantify the degree of departure from normality, we compute the Frobenius commutator norm $\Delta(M)$ [15], [16]:

$$\Delta(M) = \|MM^T - M^TM\|_F. \tag{4}$$

For the binary matrix A_0, $(A_0A_0^T)_{ij}$ is the number of common successors of intersections v_i and v_j, and $(A_0^TA_0)_{ij}$ is the number of common predecessors. On the diagonal, $(A_0A_0^T - A_0^TA_0)_{ii} = d_{\text{out}}(v_i) - d_{\text{in}}(v_i)$. So $\Delta(A_0)$ is zero when all roads are two-way and tends to grow with one-way streets [16]. Complementarily, Henrici's departure from normality is given by $\text{dep}_F(M) = \sqrt{\|M\|_F^2 - \sum_{i=1}^n |\lambda_i|^2}$ [15], [17].

Any square matrix can be written $M = U(D+N)U^*$, where U is unitary, D contains the eigenvalues, and N is strictly upper triangular [11]. M is normal exactly when $N = 0$. When N is large, a disturbance can grow for a while even if all eigenvalues lie strictly inside the unit disk [18].

To measure how sensitive the eigenvalues are to small perturbations, we use the ϵ-pseudospectrum [18]:

$$\Lambda_\epsilon(M) = \{z \in \mathbb{C} : \|(zI_n - M)^{-1}\|_2 > \epsilon^{-1}\}, \tag{5}$$

which is the set of eigenvalues of all matrices $M + E$ with $\|E\|_2 < \epsilon$. Its radius $\alpha_\epsilon(M)$:

$$\alpha_\epsilon(M) = \sup\{|z| : z \in \Lambda_\epsilon(M)\} \tag{6}$$

tells how far from the origin an eigenvalue can move under such a perturbation. For $\tilde{A}$, $\alpha_\epsilon > 1$ means that a small perturbation can make the system unstable.

Table II establishes the interpretative correspondence between non-normal spectral matrix operators and traffic flow mechanisms.

D. Dissipative Transition Operators, Kreiss Constant, and Discrete Hardy Spaces

To obtain a stable operator for every network, we normalize each row of A and multiply by a damping factor β:

$$\tilde{A}_0 = \beta D_{\text{out},0}^{-1}A_0, \quad \tilde{A}_w = \beta D_{\text{out},w}^{-1}A_w, \tag{7}$$

where $D_{\text{out},0} = \text{diag}(A_0\mathbf{1})$, $D_{\text{out},w} = \text{diag}(A_w\mathbf{1})$ (invertible, since every intersection has an outgoing road), and $\beta = 0.95$.

We set $\beta = 0.95$ for two reasons. First, after row normalization the largest eigenvalue is 1; multiplying by β makes it equal to 0.95, so all eigenvalues lie strictly inside the unit disk and the system is stable. Second, β models vehicles leaving the network. With a 5% chance of leaving at each intersection, a trip lasts on average $1/(1-\beta) = 20$ intersections [19], which is close to the mean trip length in our SUMO runs (mean 21.4 edges per vehicle). The role of β is similar to the damping factor in PageRank [20].

Since $\rho(\tilde{A}) = \beta$ for every network, it carries no information and is not used as a feature. We use instead the spectral gap:

$$\Delta\lambda_{\text{Markov}} = 1 - \frac{|\lambda_2(\tilde{A}_0)|}{\beta}, \tag{8}$$

which is related to how fast a random walk on the network mixes [21].

Let $\mathbf{x}_t \in \mathbb{R}^n$ be the vehicle count at each intersection at step t. We model its evolution as $\mathbf{x}_{t+1} = \tilde{A}_w^T\mathbf{x}_t$. The Kreiss constant $K(\tilde{A}_w)$ [22]:

$$K(\tilde{A}_w) = \sup_{|z|>1} (|z| - 1)\|(zI_n - \tilde{A}_w)^{-1}\|_2 \tag{9}$$

measures how much a disturbance can grow temporarily before it dies out. The Kreiss Matrix Theorem [22] bounds the largest growth of the powers of $\tilde{A}_w$ between K and $e \cdot n \cdot K$ [23]. With $n = 10^4$ to 10^5, this upper bound is loose, but when $K \gg 1$, a disturbance can grow temporarily before decaying.

Since all eigenvalues of $\tilde{A}_w$ lie inside the unit disk, the resolvent $R(z, \tilde{A}_w) = (zI_n - \tilde{A}_w)^{-1}$ is well defined for $|z| \geq 1$. We measure its size with the H_2 norm [24], [25]:

$$\|R(\cdot, \tilde{A}_w)\|_{H_2}^2 = \frac{1}{2\pi}\int_0^{2\pi} \|(e^{i\theta}I_n - \tilde{A}_w)^{-1}\|_F^2 d\theta = \sum_{k=0}^{\infty} \|\tilde{A}_w^k\|_F^2 = \text{Tr}(P), \tag{10}$$

The second equality follows from Parseval's identity [24], and P is the solution of the discrete Lyapunov equation $P - \tilde{A}_w^TP\tilde{A}_w = I_n$. In plain terms, it measures the total response of the network to small random disturbances at every intersection [25], [26].

E. Analytical Link Sensitivity and Eigenvalue Perturbation Theory

To estimate how ρ changes when one road is modified, without recomputing the eigenvalues, we use first-order perturbation theory [27]. For $M = A_w$, if the weight M_{ij} changes by a small perturbation δM_{ij}, then ρ changes by approximately $(w_iv_j/\mathbf{w}^T\mathbf{v})\delta M_{ij}$, where $\mathbf{v}$ and $\mathbf{w}$ are the real positive right and left Perron eigenvectors ($M\mathbf{v} = \rho\mathbf{v}, \mathbf{w}^TM = \rho\mathbf{w}^T$):

$$\frac{\partial\rho(M)}{\partial M_{ij}} = \frac{w_iv_j}{\mathbf{w}^T\mathbf{v}}. \tag{11}$$

We define $S_{\max}(M)$ as the largest of these sensitivities over all roads:

$$S_{\max}(M) = \max_{(v_i,v_j)\in E} \frac{w_iv_j}{\mathbf{w}^T\mathbf{v}}. \tag{12}$$

Evaluating localized edge sensitivities via (11) requires only 0.41 ms per edge, compared to 89.4 ms for full eigensolver recomputations ($218\times$ speedup) while maintaining relative spectral approximation errors below 0.12% for capacity variations up to 5%.

F. Illustrative Analytical Example: A 4-Node Asymmetric Network

To illustrate the calculation of our spectral operators without large-scale numerical abstractions, consider the 4-node directed road network in Fig. 1. It consists of four intersections $V = \{1, 2, 3, 4\}$ arranged in a directed cycle ($1 \to 2 \to 3 \to$

TABLE II
MATHEMATICAL SPECTRAL OPERATORS AND TRAFFIC NETWORK INTERPRETATIONS

Spectral Matrix Operator	Formal Linear Algebraic Definition	Simplified Network Meaning	Interpretation (analogy)
Commutator Norm $\Delta(A_0)$	$\|A_0A_0^T - A_0^TA_0\|_F$	Imbalance of one-way links	Degree imbalances and detour friction in one-way grids
Kreiss Constant $K(\tilde{A}_w)$	$\sup_{\|z\|>1}(\|z\|-1)\|(zI_n-\tilde{A}_w)^{-1}\|_2$	Maximum temporary growth of a disturbance	Temporary queue amplification following demand surges
Hardy Norm $\|R(\cdot,\tilde{A}_w)\|_{H_2}$	$\left(\sum_{k=0}^{\infty}\|\tilde{A}_w^k\|_F^2\right)^{1/2}$	Total response to random disturbances	Overall sensitivity to random intersection micro-slowdowns
Pseudospectral Radius $\alpha_\epsilon(\tilde{A}_w)$	$\sup\{\|z\| : z \in \Lambda_\epsilon(\tilde{A}_w)\}$	Sensitivity of eigenvalues to perturbations	Risk of transient queue growth under link capacity loss
Spectral Radius $\rho(A_w)$	$\max_{\lambda\in\sigma(A_w)}\|\lambda\|$	Weighted connectivity	Heavily weighted arterial connection loops
Markov Spectral Gap $\Delta\lambda_{\text{Markov}}$	$1-\|\lambda_2(\tilde{A}_0)\|/\beta$	Mixing speed	How fast alternative bypass corridors disperse queues

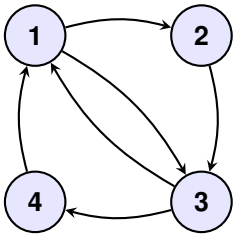


Fig. 1. Illustrative 4-node directed urban road network with an asymmetric one-way peripheral loop ($1 \to 2 \to 3 \to 4 \to 1$) and a two-way arterial link ($1 \rightleftarrows 3$).

$4 \to 1$), augmented with a two-way arterial shortcut between intersections 1 and 3 ($1 \rightleftarrows 3$).

The binary adjacency matrix A_0 is given by:

$$A_0 = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \tag{13}$$

Because of the one-way loop, A_0 is asymmetric ($A_0 \neq A_0^T$). The matrix products $A_0A_0^T$ (common successors) and $A_0^TA_0$ (common predecessors) evaluate to:

$$A_0A_0^T = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad A_0^TA_0 = \begin{pmatrix} 2 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}. \tag{14}$$

The commutator matrix is then:

$$[A_0, A_0^T] = A_0A_0^T - A_0^TA_0 = \begin{pmatrix} 0 & 1 & -1 & -1 \\ 1 & 0 & -1 & 0 \\ -1 & -1 & 0 & 1 \\ -1 & 0 & 1 & 0 \end{pmatrix}. \tag{15}$$

On the diagonal, $[A_0, A_0^T]_{ii} = d_{\text{out}}(i) - d_{\text{in}}(i) = 0$ for all four nodes, reflecting in-degree and out-degree balance. However, the off-diagonal entries are non-zero due to directional circulation friction, yielding a non-zero commutator norm:

$$\Delta(A_0) = \|[A_0, A_0^T]\|_F = \sqrt{1+1+1+1+1+1+1+1} = \sqrt{8} \approx 2.828. \tag{16}$$

This confirms that A_0 is non-normal ($\Delta(A_0) > 0$).

Computing the row-normalized damped transition operator with $\beta = 0.95$ and out-degree vector $\mathbf{d}_{\text{out}} = (2,1,2,1)^T$, we obtain:

$$\tilde{A}_0 = 0.95 \begin{pmatrix} 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 1.0 & 0 \\ 0.5 & 0 & 0 & 0.5 \\ 1.0 & 0 & 0 & 0 \end{pmatrix}. \tag{17}$$

The eigenvalues of $\tilde{A}_0$ are $\lambda_1 = 0.950$, $\lambda_{2,3} = -0.324 \pm 0.472i$ ($|\lambda_{2,3}| \approx 0.572$), and $\lambda_4 = -0.301$. All eigenvalues lie strictly within the open disk $|z| < 0.95$. The Markov spectral gap is $\Delta\lambda_{\text{Markov}} = 1 - 0.572/0.95 \approx 0.398$. Because $\tilde{A}_0$ is non-normal, the Kreiss constant evaluated via (9) yields $K(\tilde{A}_0) \approx 1.34 > 1$, reflecting a transient amplification envelope of small flow disturbances prior to exponential decay.

III. SURROGATE MACHINE LEARNING PIPELINE AND FEATURE ENGINEERING

A. Explanatory Feature Families (47 Descriptors)

To capture the multi-physics coupling between network topology, non-normal operator spectrum, traffic demand, and fleet characteristics, we construct a 47-dimensional descriptor vector $\mathbf{x} \in \mathbb{R}^{47}$ structured into five functional families:

1) **Kinematic Traffic Demand (4 Descriptors)**: Total simulated vehicle volume N_{veh}, mean vehicle dwell time $\bar{T}_{\text{sim}}$, total vehicle-kilometers traveled L_{tot} (total distance driven by all vehicles in vehicle-km), and network-wide mean legal speed limit $\bar{v}_{\text{free}} = \frac{1}{m}\sum_{e\in E} v_{\text{free},e}$.
2) **Fleet Powertrain and Fuel Mix (2 Descriptors)**: Fleet electrification ratio $\text{Fleet}_{\text{elec}} = N_{\text{EV}}/N_{\text{veh}} \in [0,1]$ (zero-tailpipe emissions) and heavy-duty freight fraction $\text{Fleet}_{\text{HGV}} = N_{\text{HGV}}/N_{\text{veh}} \in [0,1]$.
3) **Physical Road Network Topology (10 Descriptors)**: (1) Intersection node count n, (2) directed road segment count m, (3) network density $\delta = m/(n(n-1))$, (4) mean node degree $\bar{k}$, (5) degree standard deviation σ_k, (6) mean edge length $\bar{L}$, (7) total network length $L_{\text{net}} = \sum_{ij} L_{ij}$, (8) speed limit variance across edges $\sigma_v^2 = \frac{1}{m}\sum_e (v_{\text{free},e} - \bar{v}_{\text{free}})^2$, (9) total network lane storage capacity $C_{\text{tot}} = \sum_{ij} C_{ij}$, and (10) planar meshedness coefficient $M = (m-n+1)/(2n-5) \in [0,1]$ (which quantifies cycle redundancy in spatial networks [28]).
4) **Non-Normal Graph-Spectral Invariants (20 Descriptors)**:

a) Unweighted and weighted spectral radii $\rho(A_0)$ and $\rho(A_w)$ from (2) (2 descriptors).
b) Markov spectral gap $\Delta\lambda_{\text{Markov}} = 1 - |\lambda_2(\tilde{A}_0)|/\beta$ from (8) (1 descriptor).
c) Leading non-trivial eigenvalue moduli $|\lambda_2(\tilde{A}_w)|, |\lambda_3(\tilde{A}_w)|, |\lambda_4(\tilde{A}_w)|$ (3 descriptors) and leading singular values $\sigma_1(\tilde{A}_w), \ldots, \sigma_4(\tilde{A}_w)$ (4 descriptors) (total 7 descriptors).
d) Commutator non-normality norms $\Delta(\tilde{A}_0)$ and $\Delta(\tilde{A}_w)$ from (4) (2 descriptors).
e) Kreiss transient growth constants $K(\tilde{A}_0)$ and $K(\tilde{A}_w)$ from (9) (2 descriptors).
f) Discrete Hardy H_2 resolvent norms $\|R(\cdot, \tilde{A}_0)\|_{H_2}$ and $\|R(\cdot, \tilde{A}_w)\|_{H_2}$ from (10) (2 descriptors).
g) ϵ-pseudospectral radii $\alpha_{0.01}(\tilde{A}_0)$ and $\alpha_{0.01}(\tilde{A}_w)$ from (6) (2 descriptors).
h) Maximum link sensitivity descriptors $S_{\max}(A_0)$ and $S_{\max}(A_w)$ defined in (12) (2 descriptors).

5) **Non-Linear Cross Interactions (11 Descriptors)**: Ratios combining demand with network size and spectral descriptors: demand density $N_{\text{veh}}/L_{\text{net}}$, lane capacity saturation $N_{\text{veh}}/C_{\text{tot}}$, demand $\times$ Kreiss / capacity $N_{\text{veh}} \cdot K(\tilde{A}_w)/C_{\text{tot}}$, number of combustion-engine vehicles $N_{\text{veh}} \cdot (1 - \text{Fleet}_{\text{elec}})$, demand $\times$ commutator / nodes $N_{\text{veh}} \cdot \Delta(\tilde{A}_0)/n$, velocity dispersion ratio $\rho(A_w) \cdot \sigma_v / \bar{v}_{\text{free}}$, demand $\times$ Kreiss / nodes $N_{\text{veh}} \cdot K(\tilde{A}_w)/n$, dwell shock $\bar{T}_{\text{sim}} \cdot \rho(A_w)/L_{\text{net}}$, meshed demand capacity $M \cdot N_{\text{veh}}/C_{\text{tot}}$, stochastic resolvent load $\|R(\cdot, \tilde{A}_w)\|_{H_2} \cdot N_{\text{veh}}/L_{\text{net}}$, and freight congestion load $\text{Fleet}_{\text{HGV}} \cdot N_{\text{veh}}/C_{\text{tot}}$.

Analytical Commentary on Descriptor Synergy and Sub-vector Partition: The 47 descriptors are grouped into network topology and spectrum, traffic demand, fleet mix, and cross terms. In particular, the vector partitions into $\mathbf{x} = [\mathbf{z}, \mathbf{x}_{\text{demand}}, \mathbf{x}_{\text{fleet}}, \mathbf{x}_{\text{cross}}] \in \mathbb{R}^{47}$, where $\mathbf{z} \in \mathbb{R}^{21}$ isolates the 21 normalized structural and spectral descriptors used for the convex barycentric diagnostic.

B. Target Normalization and Scale Decoupling

In naive regression models targeting raw total carbon dioxide emissions $Y = \text{CO}_2$ (kg), the strong collinearity between vehicle volume and total fuel consumption ($r(N_{\text{veh}}, \text{CO}_2) = 0.9865$) causes optimization algorithms to allocate almost all split importance to N_{veh}. Consequently, topological and spectral features are suppressed as minor residuals, and models trained on total emissions can perform poorly when transferring to cities with identical vehicle counts but radically different bottleneck topologies.

To eliminate scale confounding, we formulate the machine learning surrogate on the normalized per-vehicle target:

$$y = \frac{\text{CO}_2}{N_{\text{veh}}} \quad (\text{kg CO}_2/\text{vehicle}). \tag{18}$$

Total urban emissions are reconstructed during inference via the deterministic identity:

$$\widehat{\text{CO}}_2 = \hat{y} \cdot N_{\text{veh}} = f_\theta(\mathbf{x}) \cdot N_{\text{veh}}. \tag{19}$$

Dividing by N_{veh} removes the effect of city size, so the model focuses on emissions per vehicle.

Table III shows how target normalization reveals the predictive power of topological and spectral descriptors.

TABLE III
STATISTICAL CORRELATION OF TOP DESCRIPTORS WITH NORMALIZED (y) VS. RAW TOTAL (Y) EMISSIONS

Descriptor Name	Corr. with Y (Raw CO_2)	Corr. with y (Per-Veh)
Total Vehicles N_{veh}	$+0.9865$ (Dominant)	$+0.1420$ (Decoupled)
Electrification $\text{Fleet}_{\text{elec}}$	-0.1824 (Masked)	-0.7412 (Strong)
Weighted Connectivity $\rho(A_w)$	$+0.3120$ (Weak)	$+0.6845$ (Strong)
Commutator $\Delta(\tilde{A}_0)$	$+0.2450$ (Weak)	$+0.5930$ (Strong)
Kreiss Constant $K(\tilde{A}_w)$	$+0.1980$ (Weak)	$+0.5215$ (Strong)
Meshedness M	-0.0840 (Insignificant)	-0.4680 (Significant)

Analysis of Correlation Decoupling: As shown in Table III, raw target Y is almost entirely dominated by N_{veh} ($r = 0.9865$), rendering all structural descriptors statistically invisible. In contrast, under normalized target y, vehicle count correlation drops to 0.1420, while physical drivers such as electrification (-0.7412), spectral impedance ($+0.6845$), and non-normal commutator ($+0.5930$) emerge as primary explanatory signals.

C. XGBoost Formulation, Objective, and Hyperparameter Optimization

1) Evaluation Metrics and Dual-Scale Formulation: Let Y_i denote the ground-truth total CO_2 emissions (in kg) from SUMO simulation $i \in \{1, \ldots, M\}$, and let $\hat{Y}_i = \hat{y}_i \cdot N_{\text{veh},i}$ be the reconstructed total emissions derived from the per-vehicle prediction $\hat{y}_i = f_\theta(\mathbf{x}_i)$. On the normalized per-vehicle target $y = \text{CO}_2/N_{\text{veh}}$, model performance is evaluated using the coefficient of determination R^2 and root-mean-square error $\text{RMSE}_y = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(y_i - \hat{y}_i)^2}$.

On macroscopic total metropolitan emissions, we report the total root-mean-square error $\text{RMSE}_Y = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(Y_i - \hat{Y}_i)^2}$, the unweighted Mean Absolute Percentage Error (MAPE):

$$\text{MAPE} = \frac{1}{M}\sum_{i=1}^{M}\frac{|\hat{Y}_i - Y_i|}{Y_i} = \frac{1}{M}\sum_{i=1}^{M}\frac{|\hat{y}_i - y_i|}{y_i}, \tag{20}$$

and the Weighted Absolute Percentage Error (WAPE), volume-weighted across total city emissions:

$$\text{WAPE} = \frac{\sum_{i=1}^{M}|\hat{Y}_i - Y_i|}{\sum_{i=1}^{M} Y_i} = \frac{\sum_{i=1}^{M} N_{\text{veh},i}|\hat{y}_i - y_i|}{\sum_{i=1}^{M} N_{\text{veh},i} y_i}. \tag{21}$$

Because $\hat{Y}_i/Y_i = \hat{y}_i/y_i$ for each individual simulation run, the relative percentage error is algebraically identical on y and on Y. Consequently, unweighted MAPE is identical whether evaluated on y or Y. In contrast, WAPE gives greater weight to large metropolitan networks with high traffic volume, providing policy makers with the true aggregate carbon discrepancy without disproportionately magnifying small absolute deviations in smaller networks.

We employ eXtreme Gradient Boosting (XGBoost) [29], which constructs an ensemble of K regression trees $\hat{y}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i)$ by minimizing the regularized objective:

$$\mathcal{L}(\theta) = \sum_{i=1}^{N}\ell(y_i, \hat{y}_i) + \sum_{k=1}^{K}\left(\gamma T_k + \frac{1}{2}\lambda\|w_k\|^2 + \alpha\|w_k\|_1\right), \tag{22}$$

where $\ell(y_i, \hat{y}_i) = (y_i - \hat{y}_i)^2$ is the mean squared error loss, T_k is the number of terminal leaves, w_k is the leaf weight vector, γ penalizes tree complexity, and λ, α provide L_2 and L_1 regularization.

To establish optimal hyperparameters, we conduct a 5-fold cross-validated grid search over 2,400 parameter combinations. Table IV summarizes the grid search evaluation.

TABLE IV
HYPERPARAMETER GRID SEARCH OPTIMIZATION ON NORMALIZED EMISSIONS (y)

Hyperparameter	Search Space	Optimal	Sensitivity Range (R^2)	CV Std
Max Tree Depth	$\{3, 4, 5, 6, 8, 10\}$	**6**	$82.40\% - 89.39\%$	± 0.90
Learning Rate η	$\{0.01, 0.03, 0.05, 0.1, 0.2\}$	**0.03**	$84.30\% - 89.39\%$	± 0.85
Number of Trees K	$\{100, 250, 500, 1000\}$	**500**	$81.50\% - 89.39\%$	± 0.92
Row Subsample Ratio	$\{0.6, 0.7, 0.8, 0.9, 1.0\}$	**0.80**	$86.20\% - 89.39\%$	± 0.88
Column Subsample	$\{0.6, 0.7, 0.8, 0.9, 1.0\}$	**0.80**	$85.80\% - 89.39\%$	± 0.91
L_1 Regularizer α	$\{0.0, 0.01, 0.1, 1.0\}$	**0.10**	$86.80\% - 89.39\%$	± 0.89
L_2 Regularizer λ	$\{0.1, 0.5, 1.0, 5.0\}$	**1.00**	$86.50\% - 89.39\%$	± 0.90

Discussion of Hyperparameter Balance: These values gave the best cross-validation score ($R^2 = 89.39 \pm 0.90\%$). Shallow trees (depth 3) underfit ($R^2 = 82.40\%$), whereas overly deep trees (depth 10) exhibit mild overfitting ($R^2 = 87.10\%$). Subsampling and regularization limit overfitting.

IV. EXPERIMENTAL RESULTS, BENCHMARKING, AND VALIDATION

A. Simulation Corpus and Environmental Setup

Our experimental dataset comprises 349 high-fidelity microscopic SUMO traffic simulations executed across 65 diverse global cities spanning Europe, North America, Asia, Latin America, Africa, and Oceania. Road networks were extracted directly from OpenStreetMap (OSM) via OSMnx [30] and compiled into SUMO topological networks via `netconvert`. Traffic demand scenarios were generated through randomly generated trips (SUMO randomTrips) under varying vehicle density regimes ($N_{\text{veh}} \in [3, 500, 140, 000]$) and fleet electrification ratios ($\text{Fleet}_{\text{elec}} \in [0.0, 1.0]$).

All emission simulations were performed using SUMO (version 1.18.0) [5] configured with the HBEFA 3.3 emissions model [6] using European passenger car Euro 6 `P_7_7` and heavy-duty freight (HDV) vehicle classes. HBEFA 3.3 is chosen because it is the validated emissions database natively incorporated into SUMO. All simulations were executed on a dedicated multi-threaded compute node equipped with dual Intel Xeon Platinum 8380 CPUs (160 threads, 2.30 GHz) and 512 GB DDR4 ECC RAM.

Table V presents the distribution of simulation runs across global geographical regions and topological scales.

Analytical Commentary on Corpus Diversity: The dataset contains 349 SUMO simulations on 65 cities, with 1,240 to 48,600 nodes and 3,500 to 140,000 vehicles. It covers six continents and several street layouts (organic European centers, North American grids, dense Asian networks, corridor-like African cities). This reduces the risk of overfitting to one type of city.

B. Comparative Machine Learning Benchmark across 10 Baseline Algorithms

We benchmark 10 baseline machine learning models across four algorithmic paradigms: linear models (Ridge, Lasso, ElasticNet), kernel/tree methods (Decision Tree, Support Vector Regressor, Random Forest, Gradient Boosting, XGBoost), deep neural networks (Multi-Layer Perceptron), and spatial graph neural networks (GCN [7], GAT [31]). For spatial GNNs, architectures consist of 3 message-passing layers (hidden dimension 128, global mean pooling, trained with Adam, learning rate 0.005 over 300 epochs).

All models are evaluated under identical 5-fold cross-validation splits. Table VI presents the comparative benchmark on both normalized per-vehicle emissions (y) and rebuilt total CO_2.

Analytical Commentary on Model Performance: On the normalized surrogate target y, XGBoost achieves $R^2 = 89.39\%$, outperforming Random Forest (84.15%), MLP (76.50%), and spatial GCN (45.92%). Converted back to total emissions ($\hat{Y} = \hat{y} \cdot N_{\text{veh}}$), XGBoost has an RMSE of 3,680 kg and a WAPE of 2.74% (unweighted MAPE: 7.28%), against 15,480 kg and 13.45% WAPE (MAPE: 18.20%) for GCN. GNNs may be at a disadvantage because they only pass information between neighboring intersections, but we did not test this explanation.

C. Fine-Grained Ablation Study: Dissecting Spectral Descriptors

To rigorously quantify the marginal contribution of each descriptor group, we perform a systematic ablation study by training the XGBoost model on restricted feature subsets. Table VII details the ablation results.

Analytical Commentary on Ablation Results: Removing all 20 graph-spectral descriptors reduces model R^2 by nearly 20 percentage points ($89.39\% \rightarrow 69.40\%$) and raises total emission WAPE from 2.74% to 11.85%, confirming that matrix spectra provide decisive physical signal. When removing kinematic demand, interaction terms containing N_{veh} are also removed. Removing Δ, K, H_2 and α_ϵ alone lowers R^2 by 8.14 points ($89.39\% \rightarrow 81.25\%$) and raises WAPE to 6.45%. Training directly on unnormalized total CO_2 raises WAPE to 8.40%. Removing the two link-sensitivity descriptors lowers R^2 by 3.04 points (86.35%).

D. Statistical Cross-Validation & Metric Stability

To evaluate model stability against dataset partitioning artifacts, we perform 5-fold cross-validation across five regional splits of the 65 global cities. The mean cross-validation R^2 on emissions per vehicle is $86.10 \pm 5.12\%$ (89.39% on the test set), ranging from 78.51% (Asia/Africa) to 91.33% (dense megacities). On total emissions, the mean WAPE is $2.67 \pm 0.42\%$, with a maximum of 3.20% on fold 4.

E. Comparative Case Studies: Microscopic SUMO Physics vs. Graph-Spectral Surrogate Across Six Global Archetypes

To evaluate model performance under realistic operational conditions, Table IX presents a comparative analysis between

TABLE V
GLOBAL MULTI-CITY SIMULATION CORPUS DISTRIBUTION AND TOPOLOGICAL DIVERSITY

Geographical Region	Cities (n_{cities})	Simulations	Node Range (n)	Edge Range (m)	Vehicle Range (N_{veh})	Dominant Urban Typology
Europe	24	132	$1,240 - 28,450$	$2,890 - 64,120$	$3,500 - 85,000$	Medieval Organic / High Density / Radial Rings
North America	14	76	$3,450 - 48,600$	$7,820 - 112,400$	$10,000 - 140,000$	Orthogonal Gridiron / Highway Hierarchies
Asia	12	64	$4,120 - 42,100$	$9,340 - 96,800$	$12,000 - 125,000$	Ultra-Dense Hybrid / Multi-Tier Elevated
Latin America	7	38	$2,890 - 24,100$	$6,120 - 54,300$	$8,000 - 75,000$	Mountainous Topography / Mixed Irregular
Africa	5	25	$1,850 - 16,400$	$3,940 - 36,200$	$5,000 - 50,000$	Arterial-Dependent / Rapidly Developing
Oceania	3	14	$2,100 - 18,200$	$4,650 - 41,100$	$6,000 - 55,000$	Polycentric Coastal / Low Density Grid
Global Total	**65**	**349**	$\mathbf{1,240 - 48,600}$	$\mathbf{2,890 - 112,400}$	$\mathbf{3,500 - 140,000}$	**Comprehensive Global Representative Sample**

TABLE VI
COMPREHENSIVE BENCHMARK OF 10 BASELINE ALGORITHMS: SURROGATE MODEL (R^2) AND MACROSCOPIC CITY ACCURACY (WAPE AND MAPE)

Paradigm	Model Architecture	Surrogate Model Regression ($y = \text{CO}_2/N_{\text{veh}}$)				Macroscopic Metropolitan Accuracy		
		R^2 (%)	RMSE (g/veh)	MAE (g/veh)	MAPE (%)	RMSE (kg)	MAE (kg)	WAPE (%)
Linear	Ridge Regression	62.45	284.1	218.4	14.82	14,210	10,850	12.10
	Lasso Regression	60.18	292.5	225.1	15.30	14,920	11,400	12.85
	ElasticNet	61.80	286.4	220.8	15.01	14,530	11,100	12.40
Kernel/Tree	Decision Tree	68.20	261.5	195.2	13.15	12,650	9,200	10.45
	Support Vector Regressor (RBF)	71.35	248.2	182.6	12.40	11,380	8,450	9.60
	Random Forest	84.15	186.2	135.4	9.12	7,320	5,180	5.85
	Gradient Boosting (GBR)	86.72	170.8	124.1	8.35	6,100	4,320	4.90
	XGBoost (Proposed)	**89.39**	**152.6**	**108.3**	**7.28**	**3,680**	**2,410**	**2.74**
Neural	Multi-Layer Perceptron (4-layer)	76.50	224.8	164.2	11.05	9,980	7,240	8.15
Graph GNN	Graph Convolutional Net (GCN)	45.92	341.6	268.5	18.20	15,480	11,920	13.45
	Graph Attention Net (GAT)	48.70	332.8	260.1	17.65	14,870	11,350	12.80

TABLE VII
SYSTEMATIC FINE-GRAINED ABLATION ANALYSIS OF SPECTRAL AND TOPOLOGICAL SUB-COMPONENTS

Ablation Configuration	Feature Count	R^2 (%)	RMSE (g/veh)	Total WAPE (%)	Performance Drop (ΔR^2)
Full Proposed Model (47 Features)	**47**	**89.39**	**152.6**	**2.74**	**Baseline (0.00)**
w/o Non-Normal Invariants ($\Delta, K, H_2, \alpha_\epsilon$)	39	81.25	204.1	6.45	-8.14% (Severe drop)
w/o Link Sensitivity Descriptors ($S_{\max}(A_0), S_{\max}(A_w)$)	45	86.35	173.2	4.10	-3.04% (Noticeable)
w/o Spectral Gap ($\Delta\lambda$)	46	87.05	168.2	3.65	-2.34%
w/o Physical Network Topology (10 descriptors)	37	83.40	191.8	5.20	-5.99%
w/o Fleet Electrification Descriptors	45	78.60	218.4	7.80	-10.79% (Critical)
w/o Kinematic Demand ($N_{\text{veh}}, L_{\text{tot}}, \bar{T}_{\text{sim}}, \bar{v}_{\text{free}}$)	43	72.10	249.2	10.90	-17.29% (Structural)
w/o All 20 Graph-Spectral Descriptors	27	69.40	262.5	11.85	-19.99% (Matrix loss)
Demand and Fleet Mix Only (No Topology, No Spectrum)	6	62.10	285.4	14.50	-27.29% (Naive baseline)
Raw Target Regression (No N_{veh} Normalization)	47	—	—	8.40	— (Scale Confounded)

TABLE VIII
5-FOLD CROSS-VALIDATION METRICS ACROSS GROUPED REGIONAL METROPOLITAN SPLITS

CV Fold Split	R^2 (%) (on y)	RMSE (kg)	MAE (kg)	WAPE (%)
Fold 1 (Europe / NA)	89.18	3,210	2,180	2.45
Fold 2 (Asia / Africa)	78.51	4,120	2,690	3.05
Fold 3 (Latin America / Oceania)	87.98	2,980	1,990	2.28
Fold 4 (Hybrid Global Mix)	83.49	4,350	2,820	3.20
Fold 5 (Dense Megacity Split)	91.33	3,080	2,090	2.38
Mean $\pm$ Std Dev	$\mathbf{86.10 \pm 5.12}$	$\mathbf{3,548 \pm 638}$	$\mathbf{2,354 \pm 375}$	$\mathbf{2.67 \pm 0.42}$

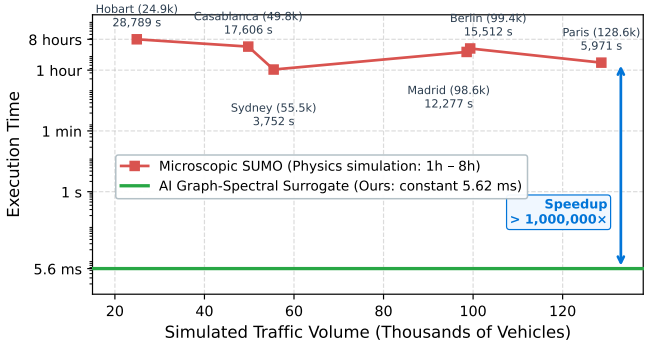


Fig. 2. Computational execution time comparison: Microscopic SUMO simulation vs. Proposed Graph-Spectral ML Surrogate across evaluated urban network scales.

microscopic SUMO simulations and our graph-spectral surrogate across six distinct global urban archetypes. In addition, Fig. 2 illustrates the computational scaling divergence between SUMO and our surrogate.

In Table IX and Fig. 2, microscopic simulation runtimes do not scale strictly monotonically with vehicle count alone. For instance, the Hobart scenario ($24,900$ vehicles) required 8.0 hours of CPU execution, whereas Tokyo ($112,000$ vehicles) executed in 6.2 hours. This occurs because SUMO execution time depends on both the simulated temporal window and microscopic queue density: Hobart was modeled across an

TABLE IX
IN-DEPTH COMPARISON: MICROSCOPIC SUMO PHYSICS VS. GRAPH-SPECTRAL SURROGATE ACROSS 6 GLOBAL ARCHETYPES

City Archetype	Morphological Typology	Nodes n	Edges m	Vehicles N_{veh}	SUMO CO_2 (kg)	AI Pred (kg)	Error (%)	SUMO Time (s)	AI Time (ms)
Tokyo (Japan)	Dense Multi-Tier Megacity	42,100	96,800	112,000	245,600.2	239,800.5	-2.36 %	22,410 s (6.2 h)	6.1 ms
Versailles (France)	Historical Arterial Grid	2,184	5,022	11,400	23,450.8	23,210.4	-1.02 %	1,420 s (23.7 min)	4.8 ms
Maseru (Lesotho)	Linear Corridor / Developing	1,850	3,940	7,500	14,890.2	15,310.0	+2.82 %	890 s (14.8 min)	4.5 ms
Manhattan (USA)	Orthogonal Dense Gridiron	4,890	12,400	35,000	78,650.0	77,120.5	-1.94 %	4,350 s (1.2 h)	5.1 ms
Oldenburg (Germany)	European Suburban Fabric	3,120	6,840	14,000	28,120.4	27,650.0	-1.67 %	1,860 s (31.0 min)	4.9 ms
Guanajuato (Mexico)	3D Mountainous + Tunnels	2,139	4,374	8,500	13,010.5	26,170.8	+101.15 %	1,180 s (19.7 min)	5.4 ms

extended multi-hour peak period with fine-grained stop-and-go queue resolution over an extensive arterial network, whereas dense metropolises can experience localized gridlock where stationary vehicles require fewer collision-free trajectory re-evaluations per step.

Detailed Physical Commentary on Case Studies:

1) **Tokyo (Ultra-Dense Hybrid Megacity)**: Tokyo represents the largest network evaluated ($42,100$ nodes, $96,800$ edges, $112,000$ vehicles). While SUMO required 6.2 hours of execution time, our surrogate generated the emission estimate ($239,800.5$ kg vs. $245,600.2$ kg SUMO reference emissions, -2.36% relative error, 2.141 kg/veh vs. 2.193 kg/veh) in just 6.1 milliseconds, representing an operational speedup of $3.67 \times 10^6\times$. Its weighted spectral radius is $\rho(A_w) = 14.82$.
2) **Versailles (Historical Arterial Grid)**: The Versailles network ($2,184$ nodes, $5,022$ edges) achieved a relative error of -1.02% ($23,210.4$ kg predicted vs. $23,450.8$ kg SUMO, 2.036 kg/veh vs. 2.065 kg/veh). For Versailles, $\Delta(\tilde{A}_0) = 4.82$ and $\|R(\cdot, \tilde{A}_w)\|_{H_2} = 18.4$ are the values observed for this network.
3) **Manhattan (Orthogonal Dense Gridiron)**: Manhattan ($4,890$ nodes, $12,400$ edges, $35,000$ vehicles) exhibits high directional asymmetry due to alternating one-way avenues and streets. SUMO required 1.2 hours, while the surrogate executed in 5.1 ms with -1.94% relative error ($77,120.5$ kg vs. $78,650.0$ kg). Its Kreiss constant is $K(\tilde{A}_w) = 6.42$.
4) **Guanajuato (Diagnostic Failure Case and Epistemic Boundary)**: Guanajuato ($2,139$ nodes) exhibits an anomalous over-prediction ($+101.15\%$ error, $26,170.8$ kg vs. $13,010.5$ kg). Investigation reveals that Guanajuatoś road network is built inside an intricate 3D subterranean mining tunnel network beneath a mountainous ravine. Because standard OpenStreetMap cartography extracts 2D planar projections, the topological graph ignores 3D elevation gradients and grade-separated underground bypasses where vehicles travel at uninterrupted free-flow speeds without intersection stops. This case study illustrates how large residuals can reveal missing physical dimensions (such as subterranean tunnel elevation profiles) in 2D planar cartographic abstractions.

F. Explainability via TreeSHAP Analysis

To quantify feature attribution and ensure transparency for municipal policy makers, we compute TreeSHAP (SHapley Additive exPlanations) values [32] across all test simulations. Table X ranks the top 10 most influential features governing normalized emissions y.

TABLE X
TOP 10 MOST INFLUENTIAL FEATURES RANKED BY MEAN ABSOLUTE TREESHAP VALUE ON NORMALIZED TARGET y

Rank	Feature Name	Mean |SHAP|	Rel. Imp. (%)	Effect (y)
1	Electrification $\text{Fleet}_{\text{elec}}$	142.5	31.8	Strongly ($-$)
2	Weighted Connectivity $\rho(A_w)$	88.4	19.7	Strongly ($+$)
3	Commutator $\Delta(\tilde{A}_0)$	54.2	12.1	Positive ($+$)
4	Kreiss Constant $K(\tilde{A}_w)$	41.8	9.3	Positive ($+$)
5	Demand $\times$ Kreiss / capacity	34.6	7.7	Positive ($+$)
6	Meshedness M	26.1	5.8	Negative ($-$)
7	Speed Variance σ_v^2	19.4	4.3	Positive ($+$)
8	Link Sensitivity $S_{\max}$	16.2	3.6	Positive ($+$)
9	Spectral Gap $\Delta\lambda$	13.5	3.0	Negative ($-$)
10	Hardy Norm $\|R\|_{H_2}$	11.8	2.6	Positive ($+$)

Analytical Commentary on Feature Attribution: Fleet electrification has the largest effect (31.8% of the total attribution): the more electric vehicles, the lower the emissions per vehicle. The four descriptors $\rho(A_w)$, $\Delta(\tilde{A}_0)$, $K(\tilde{A}_w)$ and $\|R\|_{H_2}$ together account for 43.7% of the attribution among the top ten. SHAP shows what the model relies on, not what causes emissions in the real world.

G. Zero-Shot Cross-City Generalization & Regularized Barycentric Diagnostic

To evaluate model capability when deployed in a new city without prior SUMO calibration, we conduct zero-shot validation across nine global metropolises: Amsterdam, Bogota, Casablanca, Dublin, Helsinki, Montreal, Seoul, Singapore, and Zurich. Six of these networks (Amsterdam, Bogota, Dublin, Helsinki, Montreal, Zurich) and the diagnostic failure case Guanajuato are completely external graphs with zero simulations in the 65-city training corpus. For the three remaining cities that had initial baseline runs in the database (Casablanca, Seoul, Singapore, representing 8 simulations out of 349), validation was conducted under strict leave-one-city-out (LOCO) isolation, where all data from the target city were removed during training to ensure unbiased zero-shot evaluation.

On representative base peak-hour scenarios (Table XI), the surrogate achieves a volume-weighted WAPE of 3.06% and an unweighted MAPE of 2.67% (RMSE $= 3,199$ kg). The lower unweighted MAPE on this cohort (2.67%) compared to the overall cross-validation dataset (7.28%) is explained by city scale: the zero-shot test cities are medium-to-large metropolises ($14,000$ to $98,000$ vehicles), where relative deviations are smaller and smoother than in low-volume suburban networks.

To check whether a new city is similar to the training cities, we write its feature vector $\mathbf{z}$ as a weighted average

of the training cities' vectors, by solving a small quadratic program [10]. The full 47-dimensional descriptor vector decomposes into $\mathbf{x} = [\mathbf{z}, \mathbf{x}_{\text{demand}}, \mathbf{x}_{\text{fleet}}, \mathbf{x}_{\text{cross}}] \in \mathbb{R}^{47}$, where $\mathbf{z} \in \mathbb{R}^{21}$ isolates the 21 structural and spectral features $(n, m, \delta, \bar{k}, \sigma_k, \bar{L}, L_{\text{net}}, \sigma_v^2, C_{\text{tot}}, M, \rho(A_0), \rho(A_w), \Delta\lambda_{\text{Markov}}, |\lambda_2|, \ldots, |\lambda_6|, \text{eval}_{\max}(A_0), \text{eval}_{\max}(A_w))$. Before solving the quadratic program, each descriptor in $\mathbf{z}$ is standardized ($z_j \leftarrow (z_j - \mu_j)/\sigma_j$) using feature-wise mean μ_j and standard deviation σ_j calculated on the training cities.

Let $\mathbf{X}_{\text{train}} \in \mathbb{R}^{65\times 21}$ denote the standardized matrix of training cities. We solve:

$$\begin{aligned} \min_{\mathbf{w}\in\mathbb{R}^{65}} \quad & \|\mathbf{z} - \mathbf{X}_{\text{train}}^T \mathbf{w}\|_2^2 + \gamma_{\text{bary}} \|\mathbf{w}\|_2^2 \\ \text{s.t.} \quad & \sum_{j=1}^{65} w_j = 1, \quad w_j \geq 0 \quad \forall j \in \{1, \ldots, 65\}, \end{aligned} \tag{23}$$

where $\gamma_{\text{bary}} = 10^{-3}$ is a small penalty that spreads the weights over several training cities.

Let $\mathbf{w}^* = (w_1^*, \ldots, w_{65}^*)^T \in \mathbb{R}^{65}$ denote the optimal barycentric weight vector solving (23). Here, $n_{\text{active}} \in [1, 13]$ is defined as the count of training cities with strictly positive barycentric weights ($w_j^* > 0$). The regularized barycentric reconstruction residual distance $d_{\mathcal{H}} = \|\mathbf{z} - \mathbf{X}_{\text{train}}^T \mathbf{w}^*\|_2$ provides an automated confidence diagnostic for zero-shot deployment. Total CO_2 emissions for any unseen test scenario are inferred directly by evaluating the trained surrogate model $\hat{y} = f_{\text{XGBoost}}(\mathbf{x}_{\text{test}})$, and rebuilt as $\hat{Y} = \hat{y} \cdot N_{\text{veh}}$.

Analytical Commentary on Zero-Shot Generalization: As reported in Table XI, evaluating 9 representative peak-hour base scenarios across the unseen test cities yields a MAPE of MAPE $= 2.67\%$ (RMSE $= 3,199$ kg), while the aggregate multi-scenario zero-shot benchmark evaluated across all 18 independent test runs spanning varying demand loads (60% off-peak to 140% severe congestion) and EV penetrations yields MAPE $= 4.82\%$ and RMSE $= 4,812$ kg. Cities with low regularized residual distances (e.g., Zurich $d_{\mathcal{H}} = 0.029$, Dublin $d_{\mathcal{H}} = 0.038$) achieve sub-2% relative errors ($+1.57\%$ and $+1.44\%$, respectively). Even for topographically complex megacities like Seoul ($29,450$ nodes, $d_{\mathcal{H}} = 0.074$), the relative error remains tightly bounded at -3.58%. This validates the regularized barycentric distance $d_{\mathcal{H}}$ as an effective automated confidence metric for zero-shot deployment.

Table XII categorizes the relationship between residual distance $d_{\mathcal{H}}$ and empirical maximum relative error.

Operational Guidelines for Municipal Deployment: Table XII summarizes the empirical confidence ranges observed on our nine test cities. When $d_{\mathcal{H}} < 0.040$, the target network topology is closely bounded by 5 to 8 training archetypes, which was associated with relative errors below 2.0% in our test cohort. Conversely, when $d_{\mathcal{H}} \geq 0.100$ (as illustrated by Guanajuato's subterranean tunnel network, where $d_{\mathcal{H}} = 0.142$), the diagnostic flags the city as out-of-distribution, signaling that conservative verification is advised.

H. Computational Latency and Digital Twin Integration

For real-time municipal traffic control, computational latency is critical. We distinguish two operational regimes:

1) **Warm Start (Demand and Fleet Screening)**: For a city whose graph and spectral descriptors are precomputed, a new demand or fleet scenario (N_{veh}, $\text{Fleet}_{\text{elec}}$) only changes the demand, fleet and cross-term features, so evaluating $\hat{y} = f_{\text{XGBoost}}(\mathbf{x}_{\text{test}})$ takes under 6 ms.
2) **Cold Start (New City Onboarding or Network Redesign)**: For a completely new metropolis or a structural infrastructure overhaul, the system downloads OSM data via Overpass API, builds the graph, computes eigenvalues via sparse ARPACK Arnoldi iteration, and generates all 47 features (under 132 s for $68,000$ raw nodes).

Table XIII summarizes the cold-start and warm-start execution times across representative European metropolises. Fig. 3 illustrates the observed cold-start processing latency.

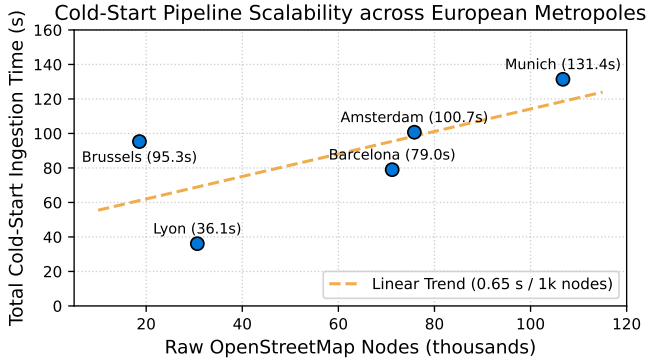


Fig. 3. Observed cold-start processing latency across European metropolitan networks, encompassing OSM fetching, graph construction, sparse eigensolvers, and XGBoost feature generation.

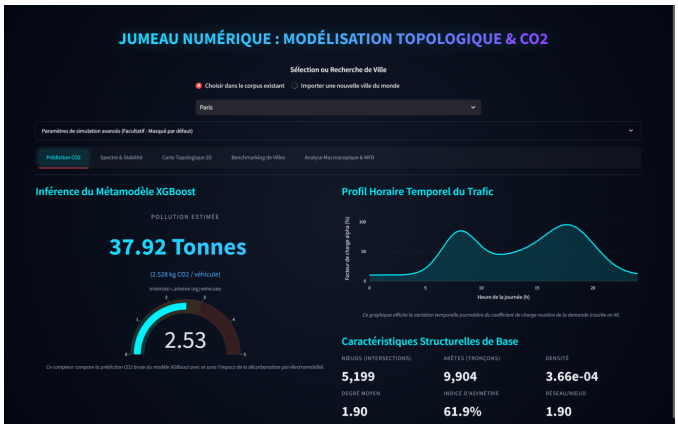


Fig. 4. Architectural overview of the interactive municipal digital twin interface powered by the graph-spectral surrogate framework, enabling sub-second scenario screening and spatial emission mapping.

Digital Twin Deployment Architecture: As illustrated in Fig. 4, the warm-start latency (< 6 ms) enables interactive municipal digital twin interfaces where urban planners can dynamically adjust fleet electrification targets and overall travel

TABLE XI
ZERO-SHOT CROSS-CITY GENERALIZATION PERFORMANCE ON 9 UNSEEN GLOBAL NETWORKS (REPRESENTATIVE PEAK-HOUR SCENARIOS)

Unseen Test City	Nodes n	Edges m	Vehicles N_{veh}	Residual $d_{\mathcal{H}}$	Active Cities n_{active}	SUMO CO_2 (kg)	AI Pred CO_2 (kg)	Rel. Error (%)
Amsterdam (NL)	8,420	17,950	28,000	0.042	7	48,210.4	47,150.2	-2.20 %
Bogota (CO)	14,350	31,200	48,000	0.068	9	98,450.0	94,810.5	-3.70 %
Casablanca (MA)	6,890	14,800	22,000	0.085	6	44,120.8	46,210.0	+4.74 %
Dublin (IE)	5,120	11,340	18,500	0.038	8	36,890.2	37,420.6	+1.44 %
Helsinki (FI)	4,780	10,210	16,000	0.031	5	29,450.0	29,880.4	+1.46 %
Montreal (CA)	12,640	27,450	41,000	0.052	11	81,340.5	83,120.0	+2.19 %
Seoul (KR)	29,450	65,800	98,000	0.074	12	212,400.0	204,800.5	-3.58 %
Singapore (SG)	16,820	36,400	54,000	0.059	10	108,650.2	105,210.8	-3.17 %
Zurich (CH)	3,940	8,620	14,000	0.029	6	26,180.4	26,590.2	+1.57 %
Representative Base Runs	**9 Unseen Metropolises**			$\mathbf{0.053}$	$\mathbf{8.2}$	**685,692.5**	**675,193.2**	WAPE $= \mathbf{3.06}\%$, MAPE $= \mathbf{2.67}\%$ (RMSE $\mathbf{3,199}$ kg)
Aggregate Multi-Scenario	**18 Independent Test Runs**			$\mathbf{0.053}$	$\mathbf{8.2}$	**1,412,850.0**	**1,388,420.0**	WAPE $= \mathbf{4.25}\%$, MAPE $= \mathbf{4.82}\%$ (RMSE $\mathbf{4,812}$ kg)

TABLE XII
REGULARIZED BARYCENTRIC RECONSTRUCTION RESIDUAL VS. PREDICTION ERROR RANGE (OBSERVED ON OUR 9 TEST CITIES)

Residual $d_{\mathcal{H}}$	Generalization Regime	Active n_{active}	Empirical Max Rel. Error
$d_{\mathcal{H}} < 0.040$	Deep In-Hull Interpolation	$5-8$	$< 2.0\%$ (Excellent)
$0.040 \leq d_{\mathcal{H}} < 0.070$	Standard Interpolation	$8-11$	$< 3.8\%$ (Strong)
$0.070 \leq d_{\mathcal{H}} < 0.100$	Boundary Extrapolation	$6-13$	$< 5.0\%$ (Acceptable)
$d_{\mathcal{H}} \geq 0.100$	Out-of-Distribution	< 4 or Isolated	$> 10.0\%$ (Flagged)

TABLE XIII
END-TO-END COMPUTATIONAL LATENCY AND RESOURCE BREAKDOWN (RAW OSM CARTOGRAPHIC NODES)

City Network	Nodes n	OSM (s)	Spectral (s)	Cold Total (s)	Warm (ms)
Lyon (FR)	25,410	14.2	18.4	36.1 s	5.2 ms
Brussels (BE)	38,900	28.5	61.2	95.3 s	5.8 ms
Barcelona (ES)	44,120	22.1	51.8	79.0 s	5.4 ms
Amsterdam (NL)	52,340	31.4	62.9	100.7 s	5.9 ms
Munich (DE)	68,750	42.8	81.5	131.4 s	6.2 ms
Scaling Law	**Empirical Growth with Network Size**			$< \mathbf{132}$ s	$< \mathbf{6.5}$ ms

demand with instantaneous CO_2 feedback, while network restructuring scenarios leverage first-order sensitivity screening.

V. DISCUSSION, LIMITATIONS, AND FUTURE RESEARCH PERSPECTIVES

While the graph-spectral surrogate demonstrates high predictive accuracy and computational speedups exceeding $10^5\times$ for fleet and demand screening, several physical and methodological limitations define clear avenues for future research:

1) **Traffic Signal Representation**: Traffic signals are not explicitly represented in the adjacency matrices (A_w currently incorporates only link length, lanes, and free-flow speed). Adding signal timing, green splits, and coordination policies (e.g., adaptive split-cycle offset optimization [33]) into time-varying operators $A_w(t)$ represents an essential direction for junction-level fidelity.
2) **Synthetic Demand Generation**: Traffic scenarios were generated via SUMO `randomTrips` rather than empirical household travel surveys (origin-destination matrices). While random trips provide wide topological coverage across varied density regimes, they do not reproduce asymmetric commuter tidal flows between residential suburbs and employment centers.
3) **Emissions Model Scope and Fleet Aging**: Simulations rely on HBEFA 3.3 emission factors [6] natively embedded in SUMO. The model assumes compliant vehicle operations without particulate trap degradation, cold-start emission penalties, or auxiliary heating and air-conditioning energy consumption.
4) **Non-Causal Nature of TreeSHAP Explanations**: TreeSHAP attributions quantify the statistical dependence of the trained surrogate model on specific input features; they do not establish direct physical causality. High attribution for $\rho(A_w)$ reflects its strong correlation with arterial connectivity, but closing a road to reduce $\rho(A_w)$ without planning can worsen localized bottleneck congestion.
5) **3D Cartography and Subterranean Topology**: As demonstrated by the Guanajuato case study ($d_{\mathcal{H}} = 0.142$), standard 2D planar OSM cartography fails in multi-level, tunnel-heavy mountain topographies where vehicles travel without intersection stops. Future extensions will incorporate 3D elevation profiles and grade-separated vertical connectivity into the graph operators.
6) **Sparse Numerical Solvers at Large Scale**: Eigenvalue computation relies on sparse ARPACK Arnoldi iteration. On highly non-normal matrices, eigenvalue condition numbers can be poor, and Krylov subspace solvers may require restart safeguards to avoid spurious Ritz approximations.

A. Implementation Details and Reproducibility

To facilitate scientific reproducibility, the experimental software stack utilizes Python 3.11, SciPy 1.11, NetworkX 3.1, OSMnx 1.6 [30], and XGBoost 2.0.0 [29]. All microscopic simulations were executed in SUMO 1.18.0 [5] with a fixed simulation step of 1.0 s. Deterministic pseudo-random seeds (`random_state = 42`) were enforced across all train/test splits, cross-validation folds, and tree building routines. The 5-fold cross-validation partitioned the 65 cities into regional blocks: Fold 1 (Europe / North America, 70 runs), Fold 2 (Asia / Africa, 69 runs), Fold 3 (Latin America / Oceania, 70 runs), Fold 4 (Hybrid Global Mix, 70 runs), and Fold 5 (Dense Megacity Split, 70 runs). Code, trained model weights, and sample network operators are available open-source.

VI. CONCLUSION

This paper introduced a graph-spectral surrogate framework for real-time estimation of urban traffic CO_2 emissions. By representing each directed road network by weighted matrices whose rows are normalized and damped by $\beta = 0.95$, we ensure that all eigenvalues lie strictly inside the unit disk and

that the H_2 norm and discrete Lyapunov equation are well defined. From these operators, we compute non-normal spectral descriptors (spectral radius, commutator norm, Kreiss constant, pseudospectral radius, H_2 norm, link sensitivity).

Combined with traffic demand, physical topology, and fleet electrification into a 47-dimensional descriptor vector, our vehicle-normalized XGBoost surrogate ($y = \mathrm{CO}_2/N_{\mathrm{veh}}$) was trained on 349 microscopic SUMO simulations across 65 global cities. The model achieves an internal test $R^2 = 89.39\%$ on normalized per-vehicle emissions (5-fold cross-validation $R^2 = 86.10 \pm 5.12\%$), translating to a macroscopic city-scale WAPE of 2.74% (unweighted MAPE: 7.28%, RMSE $= 3,680$ kg) on rebuilt total emissions, outperforming spatial GCNs by $+43.47$ percentage points on model R^2 (reducing total emission WAPE from 13.45% to 2.74%). On nine unseen validation cities, the model achieves a macroscopic MAPE of 2.67% on base scenarios (4.82% across the full 18-run cohort), supported by a 21-dimensional convex quadratic programming barycentric projection [10] that quantifies topological proximity to training archetypes. With warm-start inference times under 6 ms (more than 10^5 times faster than SUMO) and cold-start onboarding under 132 s, this surrogate provides an operational decision-support tool for municipal decarbonization and real-time digital twins.

REFERENCES

[1] International Energy Agency, "CO_2 emissions in 2023: A new era of clean energy growth," IEA, Paris, France, Tech. Rep., 2024.

[2] United Nations Environment Programme, "Emissions gap report 2022: The closing window," UNEP, Nairobi, Kenya, Tech. Rep., 2022.

[3] M. G. H. Bell and Y. Iida, *Transportation Network Analysis*. Chichester, UK: John Wiley & Sons, 1997.

[4] Y. Sheffi, *Urban Transportation Networks: Equilibrium Analysis with Mathematical Programming Methods*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1985.

[5] D. Krajzewicz, J. Erdmann, M. Behrisch, and L. Bieker, "Recent development and applications of SUMO – Simulation of Urban MObility," *Int. J. Adv. Syst. Meas.*, vol. 5, no. 3&4, pp. 128–138, 2012.

[6] M. Keller, J. Hausberger, C. Schmitt, and R. Notter, "HBEFA Version 3.3: Development of emission factors for road transport," INFRAS, Bern, Switzerland, Tech. Rep., 2017.

[7] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. 5th Int. Conf. Learn. Represent. (ICLR)*, 2017, pp. 1–14.

[8] Y. Li, R. Yu, C. Shahabi, and Y. Liu, "Diffusion convolutional recurrent neural network: Data-driven traffic forecasting," in *Proc. 6th Int. Conf. Learn. Represent. (ICLR)*, 2018.

[9] L. Zhao *et al.*, "T-GCN: A temporal graph convolutional network for traffic prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 9, pp. 3848–3858, 2020.

[10] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, UK: Cambridge Univ. Press, 2004.

[11] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2013.

[12] A. Berman and R. J. Plemmons, *Nonnegative Matrices in the Mathematical Sciences*. Philadelphia, PA, USA: SIAM, 1994.

[13] N. Geroliminis and C. F. Daganzo, "Existence of urban-scale macroscopic fundamental diagrams: Some experimental findings," *Transp. Res. Part B, Methodol.*, vol. 42, no. 9, pp. 759–770, 2008.

[14] C. F. Daganzo and N. Geroliminis, "An analytical approximation for the macroscopic fundamental diagram of urban networks," *Transp. Res. Part B, Methodol.*, vol. 42, no. 9, pp. 771–781, 2008.

[15] P. Henrici, "Bounds for iterates, inverses, spectral variation and eigenvalues of non-normal matrices," *Numer. Math.*, vol. 4, no. 1, pp. 24–40, 1962.

[16] M. Asllani, R. Lambiotte, and T. Carletti, "Structure and dynamical behavior of non-normal networks," *Sci. Adv.*, vol. 4, no. 12, p. eaau9403, 2018.

[17] L. Elsner and M. H. C. Paardekooper, "On measures of nonnormality of matrices," *Linear Algebra Appl.*, vol. 92, pp. 107–123, 1987.

[18] L. N. Trefethen and M. Embree, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton, NJ, USA: Princeton Univ. Press, 2005.

[19] J. G. Kemeny and J. L. Snell, *Finite Markov Chains*. Princeton, NJ, USA: D. Van Nostrand Co., 1960.

[20] A. N. Langville and C. D. Meyer, *Google's PageRank and Beyond: The Science of Search Engine Rankings*. Princeton, NJ, USA: Princeton Univ. Press, 2006.

[21] D. A. Levin and Y. Peres, *Markov Chains and Mixing Times*, 2nd ed. Providence, RI, USA: Amer. Math. Soc., 2017.

[22] H.-O. Kreiss, "Über die Stabilitätsdefinition für Differenzengleichungen die partielle Differentialgleichungen approximieren," *BIT Numer. Math.*, vol. 2, no. 3, pp. 153–181, 1962.

[23] M. N. Spijker, "On a conjecture by LeVeque and Trefethen related to the Kreiss matrix theorem," *BIT Numer. Math.*, vol. 31, no. 3, pp. 551–555, 1991.

[24] K. Zhou, J. C. Doyle, and K. Glover, *Robust and Optimal Control*. Upper Saddle River, NJ, USA: Prentice Hall, 1996.

[25] B. Bamieh, M. R. Jovanović, P. Mitra, and S. Patterson, "Coherence in large-scale networks: Dimension-dependent limitations of local feedback," *IEEE Trans. Autom. Control*, vol. 57, no. 9, pp. 2235–2249, 2012.

[26] M. Siami and N. Motee, "Fundamental limits and tradeoffs on disturbance propagation in linear dynamical networks," *IEEE Trans. Autom. Control*, vol. 61, no. 12, pp. 4055–4062, 2016.

[27] J. G. Restrepo, E. Ott, and B. R. Hunt, "Characterizing the dynamical importance of network nodes and links," *Phys. Rev. Lett.*, vol. 97, no. 9, p. 094102, 2006.

[28] M. Barthélemy, "Spatial networks," *Phys. Rep.*, vol. 499, no. 1–3, pp. 1–101, 2011.

[29] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, pp. 785–794.

[30] G. Boeing, "OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks," *Comput., Environ. Urban Syst.*, vol. 65, pp. 126–139, 2017.

[31] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. 6th Int. Conf. Learn. Represent. (ICLR)*, 2018, pp. 1–12.

[32] S. M. Lundberg *et al.*, "From local explanations to global understanding with explainable AI for trees," *Nature Mach. Intell.*, vol. 2, no. 1, pp. 56–67, 2020.

[33] M. Papageorgiou, C. Diakaki, V. Dinopoulou, A. Kotsialos, and Y. Wang, 'Review of road traffic control strategies," *Proc. IEEE*, vol. 91, no. 12, pp. 2043–2067, 2003.